

Technical Summary:

Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation

Kai-Yu Lu

1 Research Problem and Motivation

Natural Language Generation (NLG) systems can produce fluent responses while remaining unreliable, so practical deployment requires uncertainty estimation, meaning a score that predicts when a generated answer is likely incorrect. A central challenge in NLG is semantic invariance: many different surface forms can express the same meaning, so token-level uncertainty can be inflated by harmless paraphrasing. As a consequence, uncertainty computed directly over strings can conflate lexical variability with true uncertainty about the underlying meaning, which is the quantity relevant for deciding whether an answer should be trusted. The target setting is free-form question answering, where answers are short natural-language spans and correctness should be assessed at the meaning level rather than by exact string match.

2 Related Work

Standard uncertainty measures for neural models include predictive entropy and Bayesian-inspired approximations such as ensembles and MC dropout. In free-form generation, the output space is extremely large and contains many paraphrases, so directly estimating a meaningful predictive distribution over correct meanings is difficult. Simple adaptations in NLG include length-normalized predictive entropy, which reduces sensitivity to output length by normalizing token log-probabilities, but this heuristic does not address semantic equivalence between different strings.

Another line of work estimates confidence via additional supervision, such as training auxiliary predictors or prompting the model to output a truthfulness judgment. These approaches can require task-specific labels or extra training procedures. Natural Language Inference (NLI) provides tools for comparing meanings via entailment relations between sentences, motivating an entailment-based protocol to group generated strings into semantic equivalence classes before computing uncertainty.

3 Dataset Construction

No new large-scale dataset is constructed. Experiments use existing question answering datasets and small-scale human annotations to validate specific components.

- CoQA: an open-book conversational question answering setting with a supporting passage. Evaluation uses the development split with approximately 8000 questions.
- TriviaQA: a closed-book question answering setting without a supporting passage. Evaluation uses an 8000-question subset of the training split to match CoQA scale.

Table 1: Datasets and auxiliary annotations (paper-specified information only).

Item	Description	Size / Split
CoQA	Open-book conversational question answering with a supporting passage; evaluation uses the development split	~ 8000 questions (dev)
TriviaQA	Closed-book question answering; evaluation uses a subset of the training split	8000 questions (train subset)
Semantic equivalence labels	Manual labels for semantic equivalence to validate clustering	300 generations
Correctness metric validation	Manual correctness labels to validate automatic Rouge-L evaluation	200 answers

Two auxiliary human-labeled validations are reported.

- Semantic equivalence validation: 300 generated answers are manually labeled for semantic equivalence to assess clustering accuracy, reported as 92.7% on TriviaQA and 95.5% on CoQA.
- Correctness metric validation: 200 answers are manually judged and compared to an automatic Rouge-L thresholding rule; agreement accuracy is reported as 0.89 on CoQA and 0.96 on TriviaQA.

4 Query Protocol and Task Definitions

Task and success criterion

The task is uncertainty estimation for free-form question answering: given a question context, produce an uncertainty score for the generated answer such that higher uncertainty corresponds to a higher probability of being incorrect. Performance is measured by AUROC (Area Under the Receiver Operating Characteristic curve), interpreted as the probability that a randomly chosen incorrect example receives a higher uncertainty score than a randomly chosen correct example. Random ranking yields $\text{AUROC} = 0.5$.

Calibration-style scores that require probability mass over all paraphrases of a correct answer are considered intractable in free-form generation, motivating a ranking-based evaluation.

Prompting and input-output protocol

CoQA prompting uses the provided passage and optional dialogue history with the current question formatted as “Q: [question] A:” in a zero-shot setting. TriviaQA uses a 10-shot prompt formatted as multiple “Q: ... A: ...” exemplars followed by the target question, and generations are trimmed using pattern-based stopwords such as “Q:” and “Question:” to avoid continuing the prompt template.

Automatic correctness labeling

Correctness is defined by fuzzy matching using Rouge-L against the reference answer:

$$L(s, s_0) = \mathbb{I}[\text{RougeL}(s, s_0) > 0.3]. \quad (1)$$

Purpose: Defines the binary correctness label used to evaluate whether an uncertainty score separates correct from incorrect generations.

Symbols: s is a generated answer string; s_0 is a reference answer string; $\text{RougeL}(\cdot, \cdot)$ is the Rouge-L score; $\mathbb{I}[\cdot]$ is an indicator function returning 1 if the condition holds and 0 otherwise.

Intuition: The rule treats an answer as correct when it has sufficient subsequence overlap with the reference, allowing paraphrases beyond exact match.

Usage: $L(s, s_0)$ is computed per example to form the ground-truth labels for AUROC evaluation of uncertainty scores.

5 Modeling Approach

Overview

The approach estimates uncertainty over meanings rather than surface forms by (i) sampling multiple answers, (ii) clustering semantically equivalent answers into meaning classes using entailment, and (iii) computing an entropy over the induced distribution over meaning classes. The resulting score is Semantic Entropy.

Step 1: Sampling multiple answers

Given an input context x , a set of M answer sequences is sampled from the language model distribution $p(s \mid x)$. Sampling uses multinomial sampling or multinomial beam sampling. In the main analysis, $M = 10$ samples are used.

Step 2: Semantic equivalence clustering via bidirectional entailment

Semantic equivalence between two answers is operationalized via bidirectional entailment under an NLI classifier conditioned on the context: two answers are treated as equivalent if each entails the other. An NLI model (DeBERTa-large fine-tuned on MNLI) is used to classify entailment relations. To reduce redundant comparisons, clustering compares a new sample against a representative of each existing meaning class and uses transitivity as an optimization. Worst-case comparisons scale as $O(M^2)$, while the reported setting uses small M and the NLI model is substantially smaller than the generative model.

Step 3: Meaning-level probability and Semantic Entropy

Let c denote a meaning class, defined as a set of sampled sequences that are semantically equivalent under the bidirectional entailment protocol. The probability of a meaning class is defined as the sum of probabilities of its member sequences:

$$p(c \mid x) = \sum_{s \in c} p(s \mid x). \quad (2)$$

Purpose: Converts sequence-level probabilities into meaning-level probabilities by pooling paraphrases that share the same meaning.

Symbols: x is the input context; s is a generated sequence; c is a meaning class; $p(s \mid x)$ is the language model probability of sequence s .

Intuition: Multiple paraphrases expressing the same answer should contribute to one meaning-level probability mass rather than appearing as separate competing outcomes.

Usage: After clustering, Eq. (2) is applied to each meaning class to obtain a discrete distribution over meanings.

Table 2: Key experimental configuration (paper-specified information only).

Component	Setting
Generative model	OPT with 2.7B, 6.7B, 13B, 30B parameters; pre-trained; no fine-tuning
Semantic clustering model	DeBERTa-large fine-tuned on MNLI for NLI entailment classification
Uncertainty samples	$M = 10$ samples per question in the main comparisons
Prompting	CoQA: zero-shot with “Q: ... A:” format; TriviaQA: 10-shot with pattern-based trimming
Generation details	Most-likely answer uses <code>generate()</code> with <code>num_beams=5</code> , <code>do_sample=True</code> ; uncertainty sampling uses multinomial sampling with <code>num_beams=1</code> , and an alternative multinomial beam sampling setting uses <code>num_beams=5</code>
Evaluation metric	AUROC for ranking incorrect answers above correct answers by uncertainty
Hardware	80GB NVIDIA A100 GPUs
Additional hyperparameters	Not specified in the paper.

Semantic Entropy is computed as the entropy over meaning classes:

$$SE(x) = - \sum_c p(c | x) \log p(c | x). \quad (3)$$

Purpose: Produces an uncertainty score that increases when probability mass is distributed across multiple distinct meanings.

Symbols: $SE(x)$ is semantic entropy for context x ; $p(c | x)$ is the meaning-class probability from Eq. (2); the sum ranges over meaning classes produced by clustering.

Intuition: Low semantic entropy indicates concentration on one meaning even if there are many paraphrases, while high semantic entropy indicates disagreement across different meanings.

Usage: $SE(x)$ is computed per question and used as the uncertainty score evaluated by AUROC against correctness labels from Eq. (1).

Minimal end-to-end example

Consider a question with sampled answers {“Paris”, “It is Paris”, “London”}. Bidirectional entailment clustering groups the first two answers into the same meaning class and places “London” in a different class. Eq. (2) pools the probabilities of the two paraphrases into one meaning-class probability. Eq. (3) then assigns low uncertainty when most pooled probability mass concentrates on the “Paris” meaning, even though multiple surface forms are present.

6 Empirical Results

Experimental setup

Experiments use OPT language models with 2.7B, 6.7B, 13B, and 30B parameters; models are used without fine-tuning. Baselines include predictive entropy, length-normalized predictive entropy, a self-evaluation score $p(\text{True})$, and a lexical diversity proxy based on average Rouge-L overlap among sampled answers. Hardware is reported as 80GB NVIDIA A100 GPUs. Training hyperparameters are not applicable because fine-tuning is not performed in the reported experiments; additional optimizer details are not specified.

Table 3: Meaning diversity and uncertainty prediction (30B OPT, 10 samples).

Dataset	Avg. # distinct meanings		AUROC	
	Correct	Incorrect	Semantic Entropy	# meanings
CoQA	1.27	1.77	0.77	0.66
TriviaQA	1.89	3.89	0.83	0.79

Main quantitative results and interpretation

Meaning diversity separates correct from incorrect answers. For the 30B model with 10 sampled answers per question, incorrectly answered questions exhibit more distinct meanings than correctly answered questions. On CoQA, the average number of distinct meanings is 1.27 for correct cases versus 1.77 for incorrect cases; on TriviaQA, the corresponding values are 1.89 versus 3.89. This indicates that unreliability is associated with semantic disagreement across sampled generations.

Semantic Entropy improves uncertainty ranking. Semantic Entropy achieves AUROC 0.77 on CoQA and 0.83 on TriviaQA, while using only the number of distinct meanings yields AUROC 0.66 on CoQA and 0.79 on TriviaQA. The gap is larger on CoQA, suggesting that incorporating probability mass over meanings provides additional discriminative power beyond merely counting meaning clusters.

Accuracy context and task difficulty. Base answer accuracy for the 30B model is reported as 82.3% on CoQA and 50.6% on TriviaQA, indicating substantially different difficulty regimes; uncertainty estimation is evaluated under both high-accuracy and low-accuracy settings.

Ablations, robustness, and failure modes

Sampling strategy matters for uncertainty quality. Multinomial sampling produces higher diversity and higher AUROC than multinomial beam sampling: AUROC 0.758 with diversity 0.490 for multinomial sampling versus AUROC 0.735 with diversity 0.258 for multinomial beam sampling. This supports the claim that uncertainty estimation benefits from sufficiently diverse answer sets.

High-temperature generations degrade semantic clustering. At high sampling temperatures on CoQA, semantic clustering becomes unreliable due to nonsensical generations, and semantic equivalence accuracy drops to 61% at temperature $T = 1.5$. In this regime, length-normalized entropy can outperform Semantic Entropy, highlighting a weakness when sampled outputs are low quality and meaning comparisons become noisy.

Robustness to the automatic correctness metric. Alternative automatic correctness thresholds are evaluated in appendix settings for TriviaQA, and AUROC comparisons between Semantic Entropy and length-normalized entropy remain stable across several choices. Under Rouge-L > 0.3 , Semantic Entropy achieves AUROC 0.828 versus 0.802 for length-normalized entropy, with model accuracy 0.506 in that setting.

7 Summary

Uncertainty estimation for free-form NLG is reframed as uncertainty over meanings rather than strings by introducing Semantic Entropy, computed by clustering sampled generations into meaning classes using

bidirectional entailment and then measuring entropy over pooled meaning probabilities. Empirically, incorrect answers exhibit more distinct meanings than correct answers, and Semantic Entropy improves AUROC relative to sequence-level entropy baselines and a cluster-count proxy.

Innovation points versus prior work

- **Meaning-level marginalization:** Paraphrases are explicitly pooled into a meaning-level distribution via summed probabilities over semantic equivalence classes, rather than treating each surface form as a separate outcome.
- **Entailment-based equivalence protocol:** Semantic equivalence is operationalized by a bidirectional entailment test using an off-the-shelf NLI classifier conditioned on context, enabling a concrete and validated clustering procedure.
- **Unsupervised, single-model uncertainty:** The method does not require additional training, calibration labels, or ensembles, providing an off-the-shelf uncertainty estimator for existing language models.

Limitations and future directions (paper-grounded)

- **Sensitivity to generation quality at high temperature:** Semantic clustering accuracy drops to 61% at $T = 1.5$, and Semantic Entropy can be outperformed by length-normalized entropy when sampled answers become nonsensical.
- **Pairwise comparison structure:** Worst-case clustering requires $O(M^2)$ entailment checks across samples, although the reported regime uses small M and exploits transitivity to reduce comparisons.
- **Incomplete exploration of sampling hyperparameters:** Sampling hyperparameters beyond temperature, such as top- p and top- k , are noted as relevant but are not exhaustively studied.